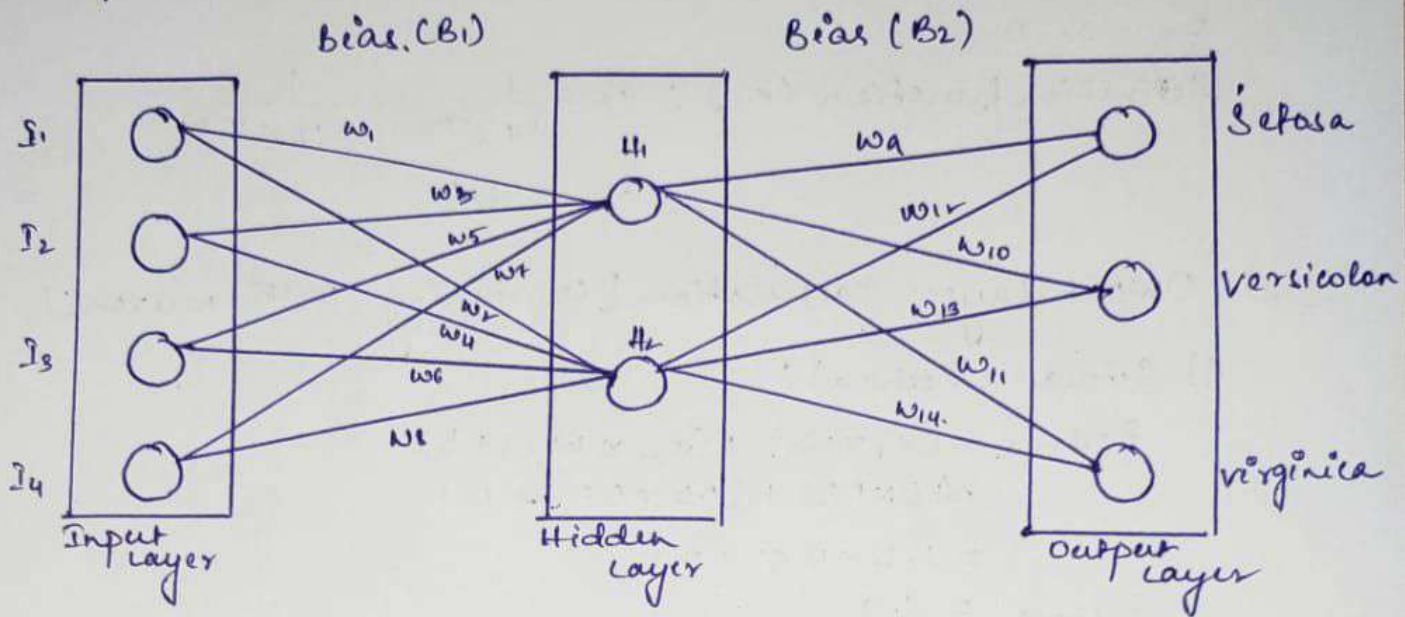


Maths Calculation Single - Hidden - Layer Neural Network.



Given Parameters:

- Input vectors: $x = [5.1, 3.2, 1.4, 0.2]$
- Hidden weight (H_1): $w_1 = 1$ $w_3 = 3$ $w_5 = 5$ $w_7 = 7$
- Hidden weight (H_2): $w_2 = 2$ $w_4 = 4$ $w_6 = 6$ $w_8 = 8$
- Output weight (from H_1): $w_9 = 1.3$ $w_{10} = 0.5$ $w_{11} = 1.2$
- Output weight (from H_2): $w_{12} = 0.5$ $w_{13} = 0.2$ $w_{14} = 0.4$
- Bias: $b_1 = 2$ $b_2 = 0.1$

Hidden Layer Calculation

A) Neuron: (H_1):

- Linear combination (z_1):

$$z_1 = x_1 w_1 + x_2 w_3 + x_3 w_5 + x_4 w_7 + b_1$$

$$z = [5.1 \times 1] + [3.2 \times 3] + [1.4 \times 5] + [0.2 \times 7] + 2$$

$$z = 5.1 + 9.6 + 7 + 1.4 + 2$$

$$z = 25.1$$

Activation function (σ_1): $\sigma_1 = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-25.1}}$

$$\sigma_1 \approx 1.0$$

8) Neuron 2 (H_2):

• Linear combination (z_2):

$$z_2 = x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + b_1$$

$$z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.6 \times 6) + (0.2 \times 8) + 2$$

$$z_2 = 35.0$$

$$\text{Activation function } (\sigma_2): \sigma_2 = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-35.0}}$$

$$\sigma_2 \approx 1.0$$

3) Output layer calculation [combining both neurons].

A) Setosa (z_{setosa}):

$$z_{\text{setosa}} = (\sigma_1 \times w_{1a}) + (\sigma_2 \times w_{1c}) + b_2$$

$$= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1$$

$$= 1.3 + 0.5 + 0.1$$

$$z_{\text{setosa}} = 1.9$$

B) Versicolour ($z_{\text{versicolour}}$):

$$z_{\text{versicolour}} = (\sigma_1 \times w_{1b}) + (\sigma_2 \times w_{1d}) + b_2$$

$$= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1$$

$$= 0.5 + 0.2 + 0.1$$

$$z_{\text{versicolour}} = 0.8$$

C) Virginica ($z_{\text{virginica}}$):

$$z_{\text{virginica}} = (\sigma_1 \times w_{1e}) + (\sigma_2 \times w_{1f}) + b_2$$

$$= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1$$

$$= 1.2 + 0.4 + 0.1$$

$$z_{\text{virginica}} = 1.7$$

4) Final Prediction

• Output vector: $[1.9, 0.8, 1.7]$

Prediction: Setosa [Highest Score 1.9].